



Complex Numbers

A complex number is any number that can be expressed in the standard form $a + bi$ or $a + jb$. The value "i" or "j" is the imaginary unit or number. It is equal to $\sqrt{-1}$.

Different Forms of Complex Numbers

1. Rectangular Form / Standard Form

$$z = a + jb$$

where:

a = real part

b = imaginary part

If the real part $a = 0$, the $z = jb$. It is called a **pure imaginary number**.

2. Trigonometric Form

$$z = r \cos \theta + jr \sin \theta$$

$$z = r \operatorname{cis} \theta$$

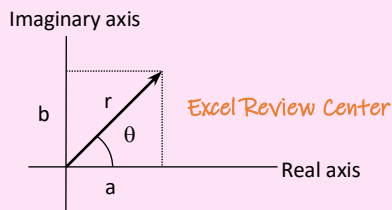
3. Polar Form

$$z = r \angle \theta$$

where:

r = modulus or absolute value

θ = argument or amplitude (in $^\circ$)



$$r = \sqrt{a^2 + b^2}$$

$$\theta = \tan^{-1} \frac{b}{a}$$

4. Exponential Form

$$z = re^{j\theta}$$

where:

θ = argument in radians

MATHEMATICAL OPERATIONS OF COMPLEX NUMBERS

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1. Addition and Subtraction of Complex Numbers

$$(a_1 + jb_1) + (a_2 + jb_2) = (a_1 + a_2) + j(b_1 + b_2)$$

$$(a_1 + jb_1) - (a_2 + jb_2) = (a_1 - a_2) + j(b_1 - b_2)$$

2. Multiplication of Complex Numbers

Use the polar form

$$(r_1 \angle \theta_1)(r_2 \angle \theta_2) = r_1 r_2 \angle (\theta_1 + \theta_2)$$

3. Division of Complex Numbers

Use the polar form

$$\frac{r_1 \angle \theta_1}{r_2 \angle \theta_2} = \frac{r_1}{r_2} \angle (\theta_1 - \theta_2)$$

4. Complex Numbers raised to exponent "n"

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Use the polar form

$$(r \angle \theta)^n = r^n \angle (n\theta)$$

5. nth root of a Complex Number

Use the polar form

$$(r \angle \theta)^{1/n} = r^{1/n} \angle \frac{\theta + k(360^\circ)}{n}$$

where:

$k = 0$, first root or principal root

$k = 1$, second root

$k = 2$, third root

$k = n - 1$, nth root

6. Logarithm of a Complex Number

Use the trigonometric form

$$\ln z = re^{j\theta} = \ln r + j\theta$$

$$\ln z = \ln r + j\theta$$

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7. Exponential and Trigonometric Functions of a Complex Number

$$e^{j\theta} = \cos \theta + j \sin \theta$$

$$e^{-j\theta} = \cos \theta - j \sin \theta$$

$$\sin \theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$

$$\cos \theta = \frac{e^{j\theta} + e^{-j\theta}}{2}$$

8. Hyperbolic Functions of Complex Number

$$\sinh \theta = \frac{e^\theta - e^{-\theta}}{2}$$

$$\cosh \theta = \frac{e^\theta + e^{-\theta}}{2}$$

Note:

$$\tanh \theta = \frac{\sinh \theta}{\cosh \theta}$$

$$\coth \theta = \frac{\cosh \theta}{\sinh \theta}$$

$$\operatorname{sech} \theta = \frac{1}{\cosh \theta}$$

$$\operatorname{csch} \theta = \frac{1}{\sinh \theta}$$

9. Inverse Trigonometric Functions of Complex Numbers

$$\arcsin z = -j \ln \left(jz \pm \sqrt{1 - z^2} \right)$$

$$\arccos z = -j \ln \left(z \pm \sqrt{z^2 - 1} \right)$$

$$\arctan z = \frac{-j}{2} \ln \left(\frac{1 + jz}{1 - jz} \right)$$

10. Inverse Hyperbolic Functions of Complex Numbers

$$\operatorname{arcsinh} z = \ln \left(z + \sqrt{z^2 + 1} \right)$$

$$\operatorname{arccosh} z = \ln \left(z \pm \sqrt{z^2 - 1} \right)$$

$$\operatorname{artanh} z = \frac{1}{2} \ln \left(\frac{1 + z}{1 - z} \right)$$

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